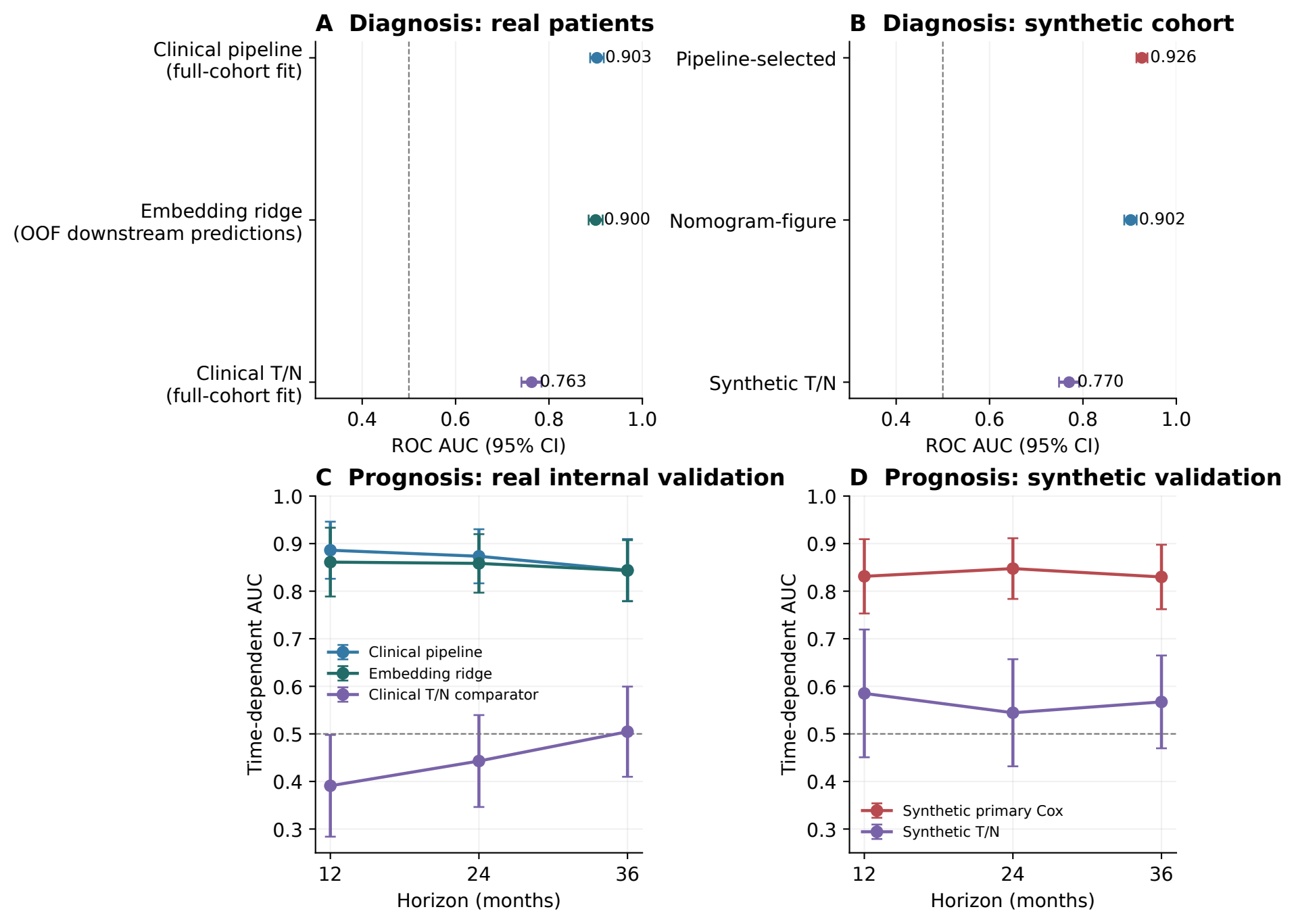
A Diagnosis: real patients
Clinical pipeline (full-cohort fit)
Embedding ridge (OOF downstream predictions)
Clinical T/N (full-cohort fit)
0.903
0.900
0.763
0.4
0.6
0.8
1.0
ROC AUC (95% CI)
B Diagnosis: synthetic cohort
Pipeline-selected
Nomogram-figure
Synthetic T/N
0.926
0.902
0.770
0.4
0.6
0.8
1.0
ROC AUC (95% CI)
C Prognosis: real internal validation
Time-dependent AUC
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
Clinical pipeline
Embedding ridge
Clinical T/N comparator
12
24
36
Horizon (months)
D Prognosis: synthetic validation
Time-dependent AUC
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
Synthetic primary Cox
Synthetic T/N
12
24
36
Horizon (months)